Algorithm: Train Coach Management using Stack and Queue

Operations: Add, Remove, Waiting Coaches, Attach & Display

1. Start the program.

2. Initialize an empty **stack** for attached train coaches.

3. Initialize an empty **queue** for waiting coaches.

4. Display the menu of operations.

5. Accept the user's choice.

6. If the user selects **Add Coach**, enter the coach number and type, then push the coach onto the stack.

7. If the user selects **Remove Coach**, remove the top coach using the **pop()** operation. Display a message if the stack is empty.

8. If the user selects **Add Waiting Coach**, enter the coach details and insert the coach into the queue using **enqueue()**.

9. If the user selects **Attach Waiting Coach**, remove the first waiting coach using **dequeue()** and attach it to the train stack.

10. If the user selects **Display Coaches**, display all attached coaches and waiting coaches.

11. Repeat the menu until the user selects **Exit**.

12. Stop the program.

— End of Algorithm —